

Two level Fractional Factorial Design 2^{K-p}

Chapter 8

- Let's suppose to run a 2^7 experimental plan
- The number of treatments are $2^7=128$.
- These data could estimate **128** model parameters:
 - 1 general mean
 - $\text{Comb}(7,1) = 7$ main effects
 - $\text{Comb}(7,2)=21$ 2 factor interactions (2fi)
 -
 - $\text{Comb}(7,7)=1$ 7 factor interaction (7fi)
- **Is it useful to build such a complex model in an explorative stage?**

Example 12.1

Box, Hunter, Hunter *Statistics for experimenter* J. Wiley

- We suspect that the **yield of a chemical process** depends on **five** factors.
- We design a **2⁵** experimental plan to understand the model

Factor	Name	Low	High
A	feeding speed	-1	1
B	catalyst	-1	1
C	stirring rate	-1	1
D	temperature	-1	1
E	concentration	-1	1

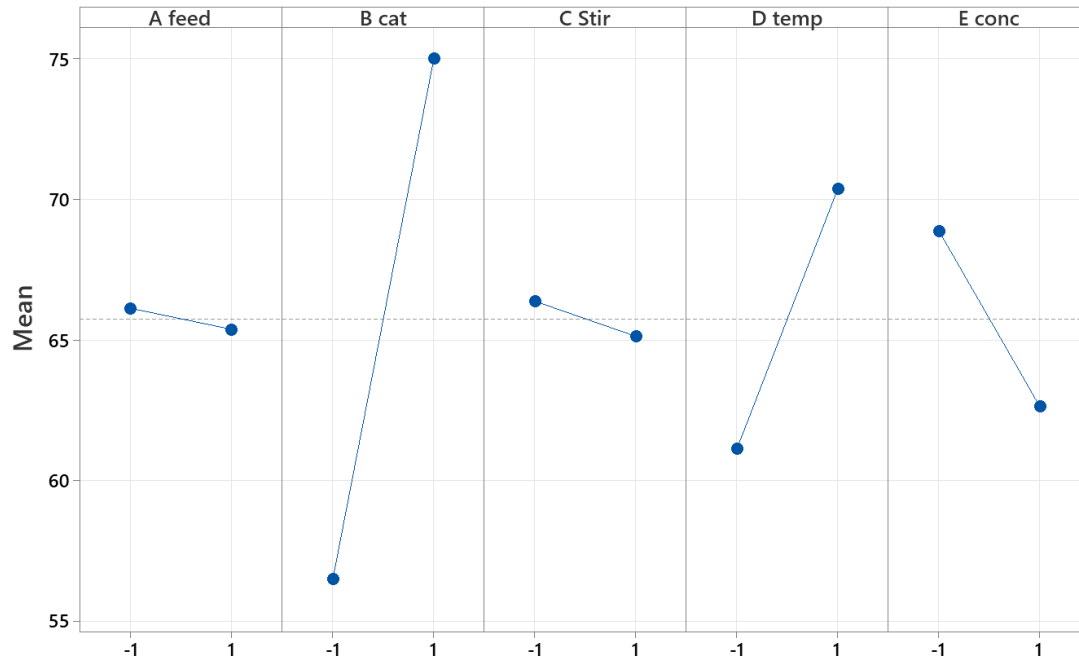
Feed cata	stirr. rate	t	conc	Yield	
-1	-1	-1	-1	61	
1	-1	-1	-1	53	
-1	1	-1	-1	63	
1	1	-1	-1	61	
-1	-1	1	-1	53	
1	-1	1	-1	56	
-1	1	1	-1	54	
1	1	1	-1	61	
-1	-1	-1	1	69	
1	-1	-1	1	61	
-1	1	-1	1	94	
1	1	-1	1	93	
-1	-1	1	1	66	
1	-1	1	1	60	
-1	1	1	1	95	
1	1	1	1	98	
-1	-1	-1	-1	1	56
1	-1	-1	-1	1	63
-1	1	-1	-1	1	70
1	1	-1	-1	1	65
-1	-1	1	-1	1	59
1	-1	1	-1	1	55
-1	1	1	-1	1	67
1	1	1	-1	1	65
-1	-1	-1	1	1	44
1	-1	-1	1	1	45
-1	1	-1	1	1	78
1	1	-1	1	1	77
-1	-1	1	1	1	49
1	-1	1	1	1	42
-1	1	1	1	1	81
1	1	1	1	1	82

Experimental data set (32 data)

Graphical analysis

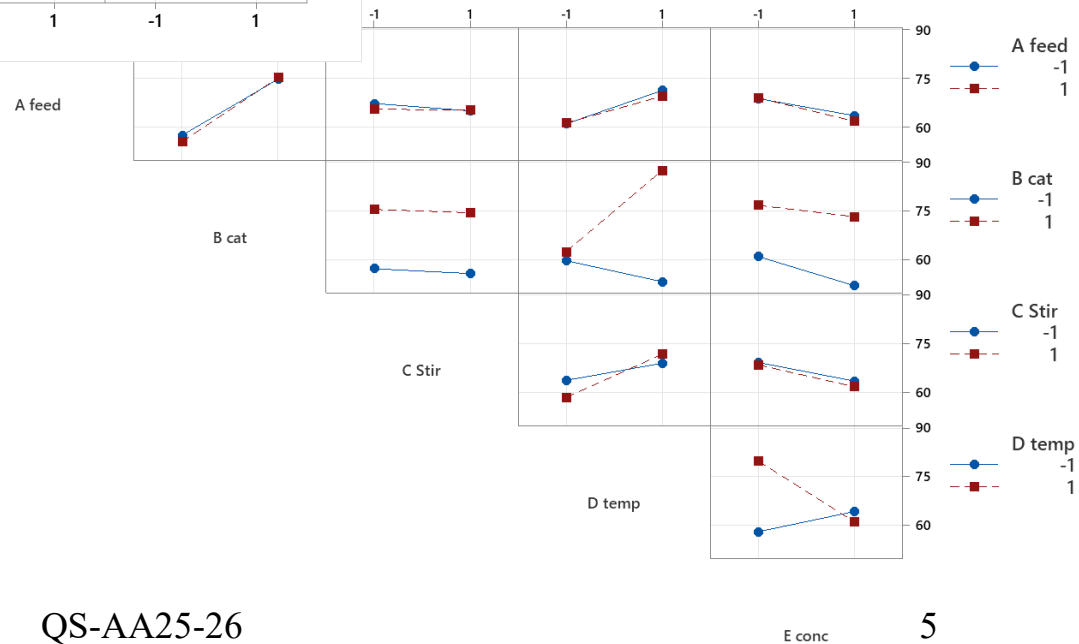
Main Effects Plot for yield

Data Means

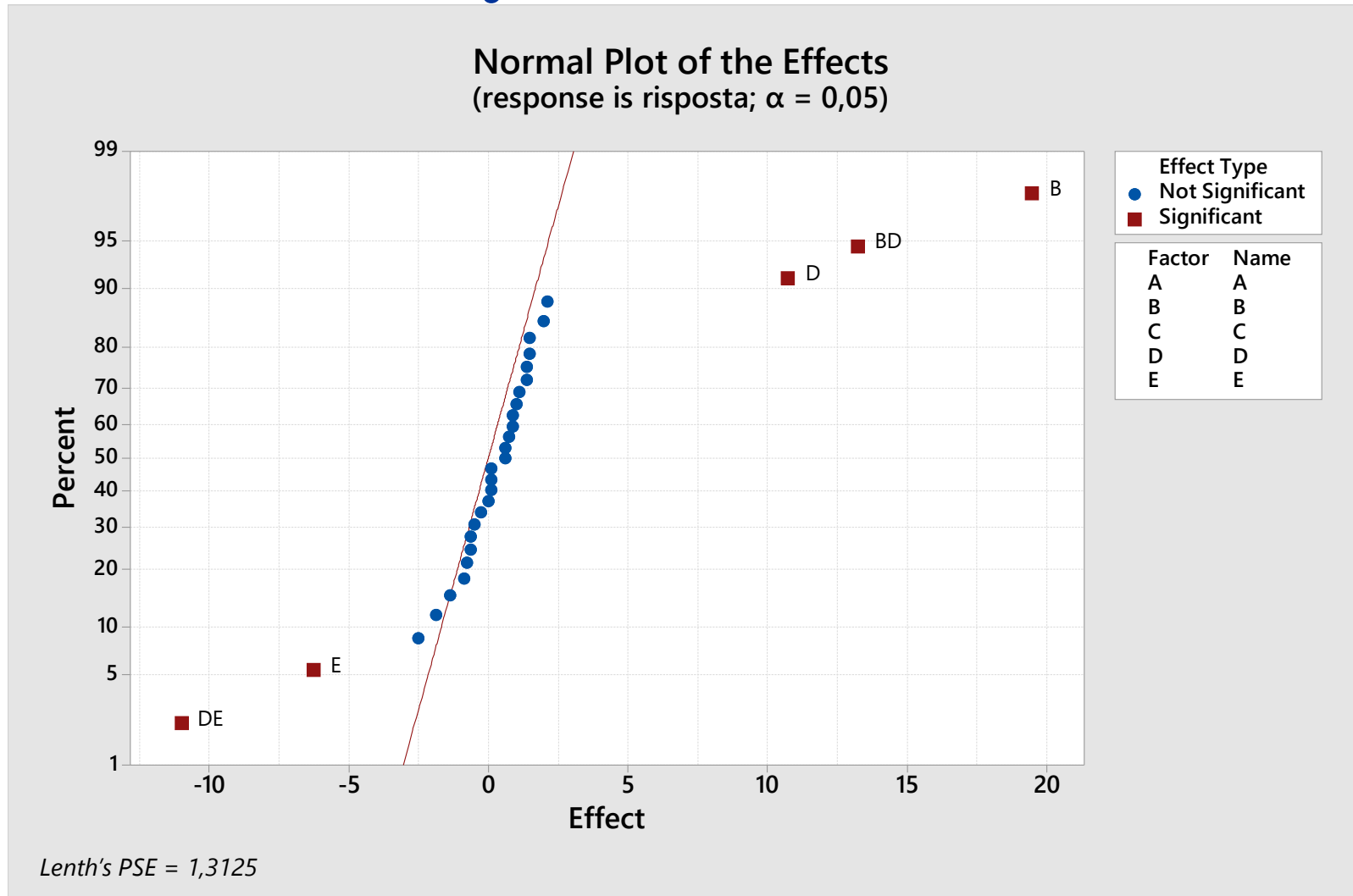


Interaction Plot for yield

Data Means



LENTH's analysis for model definition



Note JMP will give you PSE=0.656 since it considers half the effects

Quantitative analysis

Analysis of Variance

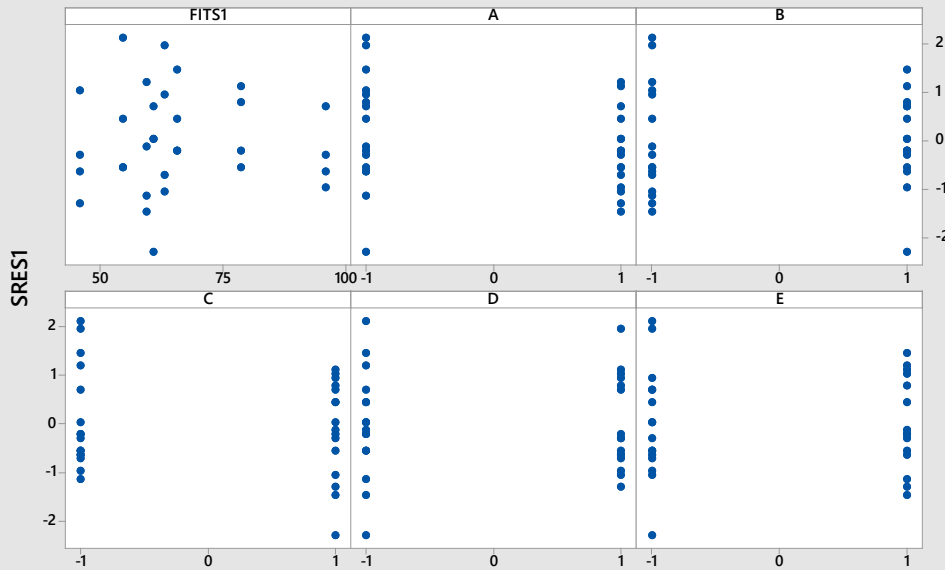
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	5	6651.5	1330.30	119.89	0.000
Linear	3	4279.0	1426.33	128.54	0.000
B	1	3042.0	3042.00	274.15	0.000
D	1	924.5	924.50	83.32	0.000
E	1	312.5	312.50	28.16	0.000
2-Way Interactions	2	2372.5	1186.25	106.91	0.000
B*D	1	1404.5	1404.50	126.58	0.000
D*E	1	968.0	968.00	87.24	0.000
Error	26	288.5	11.10		
Total	31	694.0			

Model Summary

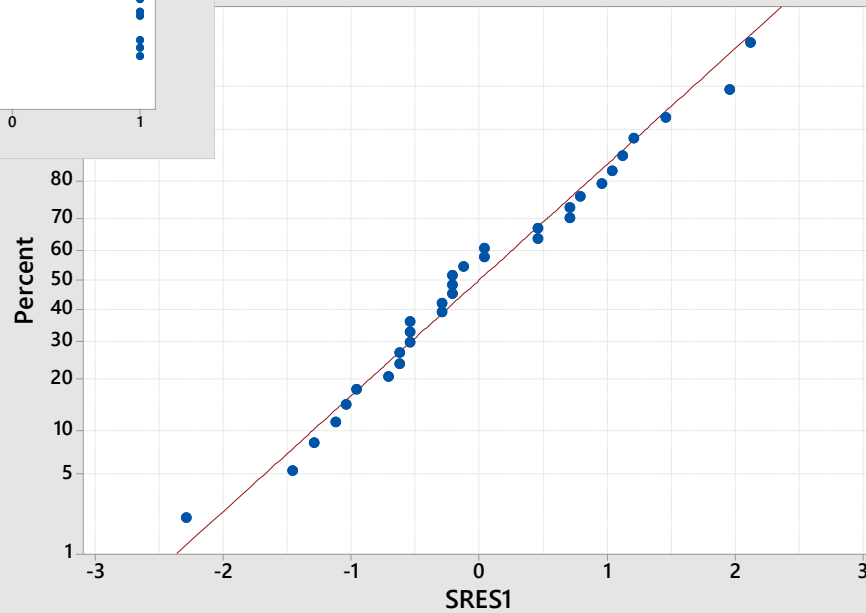
S	R-sq	R-sq(adj)
3.33109	95.84%	95.04%

Residual analysis

Scatterplot of SRES1 vs FITS1; A; B; C; D; E



Probability Plot of SRES1
Normal



Mean	-2,08167E-17
StDev	1,016
N	32
AD	0,295
P-Value	0,574

The other checks
are ok

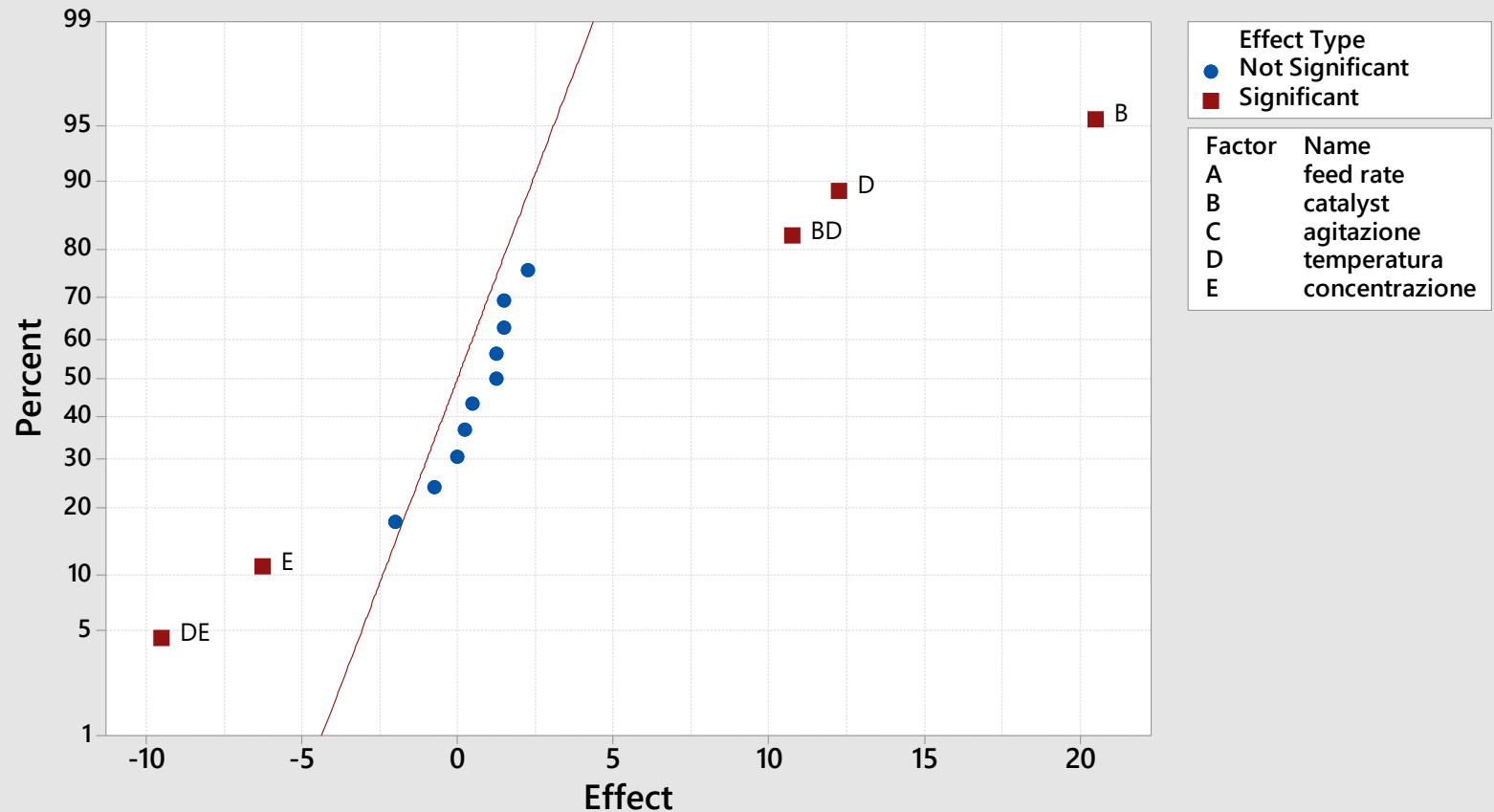
	feed rate	catalyst	StirRate	temperat	conc	Yield
1	-1	-1	-1	-1	-1	61
2	1	-1	-1	-1	-1	53
3	-1	1	-1	-1	-1	63
4	1	1	-1	-1	-1	61
5	-1	-1	1	-1	-1	53
6	1	-1	1	-1	-1	56
7	-1	1	1	-1	-1	54
8	1	1	1	-1	-1	61
9	-1	-1	-1	1	-1	69
10	1	-1	-1	1	-1	61
11	-1	1	-1	1	-1	94
12	1	1	-1	1	-1	93
13	-1	-1	1	1	-1	66
14	1	-1	1	1	-1	60
15	-1	1	1	1	-1	95
16	1	1	1	1	-1	98
17	-1	-1	-1	-1	1	56
18	1	-1	-1	-1	1	63
19	-1	1	-1	-1	1	70
20	1	1	-1	-1	1	65
21	-1	-1	1	-1	1	59
22	1	-1	1	-1	1	55
23	-1	1	1	-1	1	67
24	1	1	1	-1	1	65
25	-1	-1	-1	1	1	44
26	1	-1	-1	1	1	45
27	-1	1	-1	1	1	78
28	1	1	-1	1	1	77
29	-1	-1	1	1	1	49
30	1	-1	1	1	1	42
31	-1	1	1	1	1	81
32	1	1	1	1	1	82

Let us consider a subset
of the data set

Data of the
«simulated»
experiment (**16 data**)



Normal Plot of the Effects (response is risposta; $\alpha = 0,05$)



Lenth's PSE = 1,875

The same result !! we were a little lucky

Why are Fractional plans useful??

- The **Sparsity of effects** principle
 - You can have many factors but only **a few of them** are relevant.
 - Systems are usually dominated by main effects and low order factor interactions.
- The **Projection property**
 - Every fractional plan can be projected in a complete plan with a smaller number of factors.
- **Sequential experimentation**
 - You can add experiments to a Fractional plan to clear up a model ambiguity

The One-Half fraction of the 2^k factorial design

Note: the design has $2^K/2$ treatments/runs and is named 2^{K-1}

Let us consider the simple 2^{3-1} design

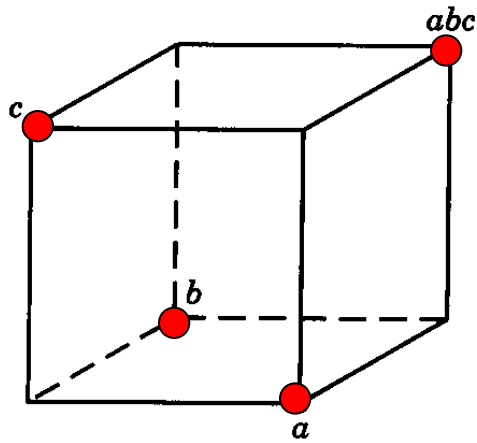
$I = ABC$ Defining relation defines one-half of the design,
the alternate fraction is defined by $I = -ABC$

Table 8-1 Plus and Minus Signs for the 2^3 Factorial Design

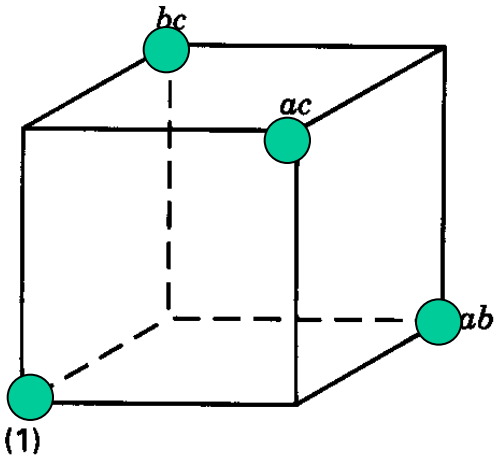
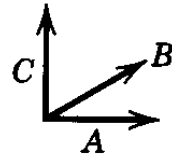
Treatment Combination	Factorial Effect							
	<i>I</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>AB</i>	<i>AC</i>	<i>BC</i>	<i>ABC</i>
<i>a</i>	+	+	−	−	−	−	+	+
<i>b</i>	+	−	+	−	−	+	−	+
<i>c</i>	+	−	−	+	+	−	−	+
<i>abc</i>	+	+	+	+	+	+	+	+
<i>ab</i>	+	+	+	−	+	−	−	−
<i>ac</i>	+	+	−	+	−	+	−	−
<i>bc</i>	+	−	+	+	−	−	+	−
(1)	+	−	−	−	+	+	+	−

Principal fraction

One-half of the 2^3 design



(a) The principal fraction, $I = +ABC$



(b) The alternate fraction, $I = -ABC$

Figure 8-1 The two one-half fractions of the 2^3 design.

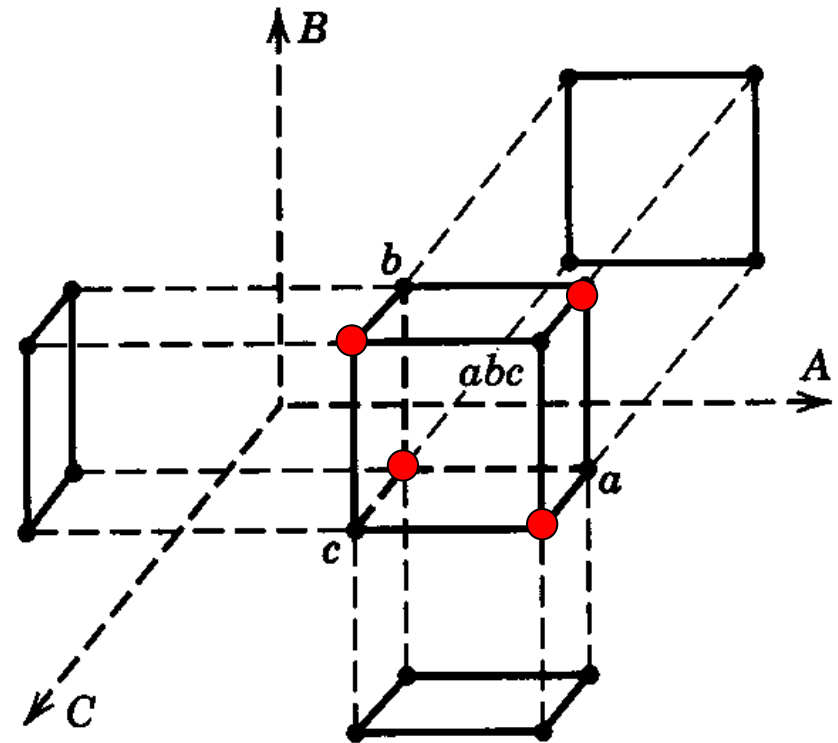


Figure 8-2 Projection of a 2^{3-1}_{III} design into three 2^2 designs.

Alias in 2^{3-1} Design

The **Defining relation** is used to find the **alias structure**

$$A = A * I = A * ABC = A^2 BC = BC$$

$$B = B * I = B * ABC = AC$$

$$C = C * I = C * ABC = AB$$

Or we can say: ***me = 2fi***

Let us compute the **Effects** (remember: they are “**aliased**”)

$$l_{[]} = \frac{\text{Reduced_Cont}_{[]}}{n2^{(k-1)-1}}$$

$$l_{[]}^* = \frac{\text{Reduced_Cont}_{[]}^*}{n2^{(k-1)-1}}$$

i.e.

$$l_A = \frac{1}{n2^{(3-1)-1}} (a - b - c + abc) = \frac{1}{2n} (a - b - c + abc)$$

$$l_A^* = \frac{1}{n2^{(3-1)-1}} (ab + ac - bc - (1)) = \frac{1}{2n} (ab + ac - bc - (1))$$

Note that

$$l_A = l_{BC} \quad l_A^* = -l_{BC}^*$$

We can write

$$EFF_A = \frac{l_A + l_A^*}{2} \quad EFF_{BC} = \frac{l_{BC} + l_{BC}^*}{2} = \frac{l_A - l_A^*}{2}$$

Hence



$$l_A = EFF_A + EFF_{BC}$$

The alternate fraction of the 2^{3-1} Design

- The defining relation is $I = -ABC$
- This implies a different Alias Structure : $A = -BC$, $B = -AC$, and $C = -AB$
- Both designs belong to the same **family** defined by

$$I = \pm ABC$$

- If we run the second-half (after the first one), we can **combine** the two groups of data and have a complete Design (or semi complete plan since we could consider a Block effect).
- This can be considered an example of **Sequential experimentation**

Design Resolution R

- **Design Resolution** is a method to characterize the fractional plans and is indicated using Roman numerals II, III, IV....
- If a design has a **Resolution R** the effects involving f factors **are not** confounded (or aliased) with effects involving **less than $R-f$** factors
- e.g. for a **Resolution V Design**
 - Main Effects (me) are not confounded with $2fi$, $3fi$ but may be confounded with $4fi$ e $5fi$
 - $2fi$ are not confounded with $2fi$ but they may be confounded with $3fi$, $4fi$ e $5fi$

- The **Resolution** must be maximized
- The **Resolution** of a 2^{k-1} Design is the **Lenght** of his **Defining Relation**.
- e.g.
 - $k=5$ $I=ABCDE$
 - Resolution V design

$$2_{IV}^{5-1}$$

Design resolution

- **Resolution III Designs** : (do **NOT** use them)
 - $me = 2fi$
 - example 2_{III}^{3-1}
- **Resolution IV Design**: (reasonable)
 - $2fi = 2fi$
 - example 2_{IV}^{4-1}
- **Resolution V Design**: (good)
 - $2fi = 3fi$
 - example 2_V^{5-1}

p = 1 (One-half Design)

k	I	Resolution	design
3	ABC	III	2_{III}^{3-1}
4	ABCD	IV	2_{IV}^{4-1}
5	ABCDE	V	2_V^{5-1}

In general, if we want to split the Design in two parts, we use the defining relation: ***I=ABCDE...K***

With **Resolution *K*** (the maximum possible resolution)

Construction of a One-Half Fraction

Let us see **another way** to build the Design.

We need: Basic Design + Design Generator

The **Basic Design** is a **Full design** using **k-1** Factors, then we add the k^{th} factor using the **Design Generator**

Table 8-2 The Two One-Half Fractions of the 2^3 Design

Run	Full 2^2 Factorial (Basic Design)		$2_{III}^{3-1}, I = ABC$			$2_{III}^{3-1}, I = -ABC$		
	A	B	A	B	C = AB	A	B	C = -AB
1	-	-	-	-	+	-	-	-
2	+	-	+	-	-	+	-	+
3	-	+	-	+	-	-	+	+
4	+	+	+	+	+	+	+	-

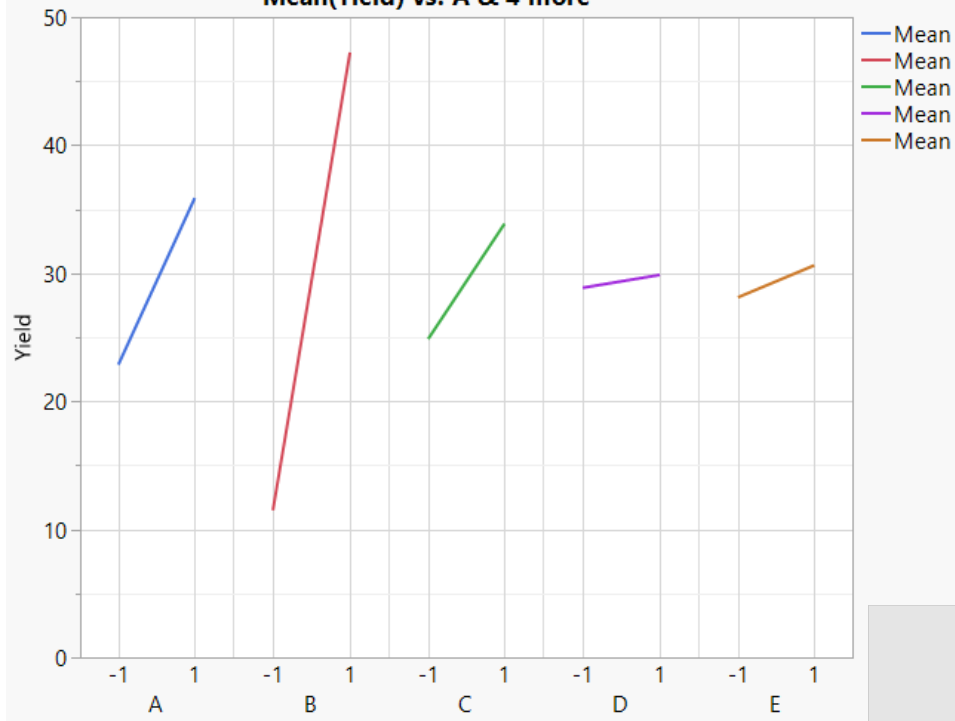
Principal fraction

Example 8.2

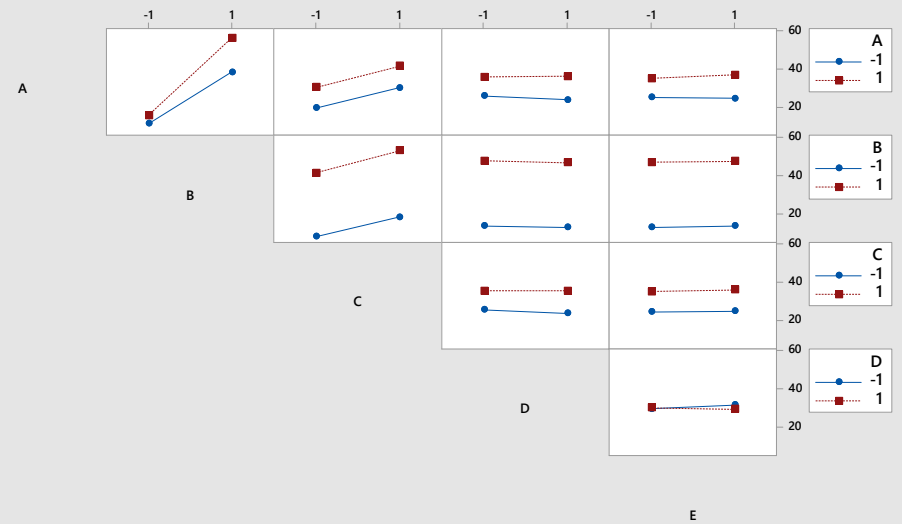
We want to investigate the effect of **5 factors** on the process **yield**. We use a **2^{5-1}** ($E=ABCD$) design and the experimental results are reported in the last column

StdOrd	RunOrd	CeterPt	Blocks	A	B	C	D	E	yield
1	1	1	1	-1	-1	-1	-1	1	8
2	3	1	1	1	-1	-1	-1	-1	9
3	7	1	1	-1	1	-1	-1	-1	34
4	9	1	1	1	1	-1	-1	1	52
5	16	1	1	-1	-1	1	-1	-1	16
6	4	1	1	1	-1	1	-1	1	22
7	15	1	1	-1	1	1	-1	1	45
8	6	1	1	1	1	1	-1	-1	60
9	12	1	1	-1	-1	-1	1	-1	6
10	14	1	1	1	-1	-1	1	1	10
11	8	1	1	-1	1	-1	1	1	30
12	5	1	1	1	1	-1	1	-1	50
13	13	1	1	-1	-1	1	1	1	15
14	11	1	1	1	-1	1	1	-1	21
15	10	1	1	-1	1	1	1	-1	44
16	2	1	1	1	1	1	1	1	63

Mean(Yield) vs. A & 4 more



Interaction Plot for Yield
Data Means



Effect Screening

The parameter estimates have equal variances.
The parameter estimates are not correlated.

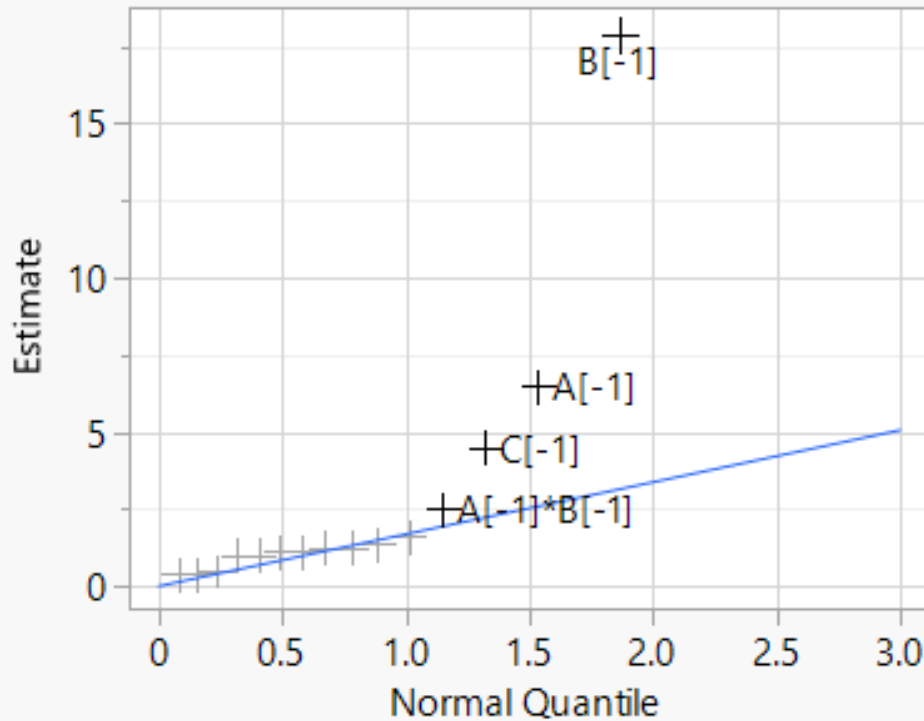
Lenth PSE

1.6875

Orthog t Test used Pseudo Standard Error

Normal Plot

Half Normal Plot ▼



Blue line has slope equal to Lenth's PSE.

Remark:

A+BCDE

B+ACDE

C+ABDE

AB+CDE

William of Ockham (1287-1347)



Ockham's Razor:

«*Pluralitas non est ponenda sine necessitate*»

or Preference for simplicity

Latin: *lex parsimoniae* "law of parsimony") is the problem-solving principle that the **simplest solution** tends to be the correct one. When presented with competing hypotheses to solve a problem, one should select the solution with the fewest assumptions

Fractional Factorial Fit:

yield versus A; B; C; D; E

Estimated Effects and Coefficients for Yield

Term	Effect	Coef
Constant		30.3125
A	11.1250	5.5625
B	33.8750	16.9375
C	10.8750	5.4375
D	-0.8750	-0.4375
E	0.6250	0.3125
A*B	6.8750	3.4375
A*C	0.3750	0.1875
A*D	1.1250	0.5625
A*E	1.1250	0.5625
B*C	0.6250	0.3125
B*D	-0.1250	-0.0625
B*E	-0.1250	-0.0625
C*D	0.8750	0.4375
C*E	0.3750	0.1875
D*E	-1.3750	-0.6875

$$\widehat{EFF}_{[]} = \frac{CONT - RED_{[]}}{n2^{(k-p)-1}}$$

$$\hat{c}_{[]} = \frac{\widehat{EFF}_{[]}}{2}$$

$$SS_{[]} = \frac{CONT - RED_{[]}^2}{n2^{k-p}} =$$
$$= n2^{(k-p)-2} \widehat{EFF}_{[]}^2 = n2^{k-p} \hat{c}_{[]}^2$$

Fractional Factorial Fit: Yield versus A; B; C

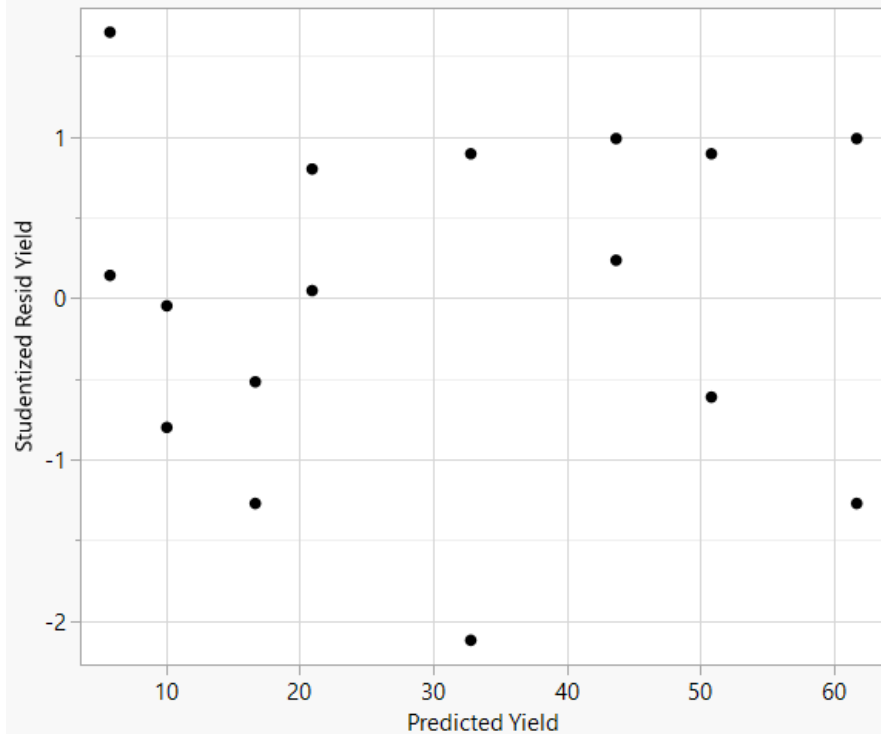
Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	4	5747.25	1436.81	560.71	0.000
Linear	3	5558.19	1852.73	723.02	0.000
A	1	495.06	495.06	193.20	0.000
B	1	4590.06	4590.06	1791.24	0.000
C	1	473.06	473.06	184.61	0.000
2-Way Interactions	1	189.06	189.06	73.78	0.000
A*B	1	189.06	189.06	73.78	0.000
Error	11	28.19	2.56		
Total	15	5775.44			

S R-sq R-sq(adj)

1.60078 99.51% **99.33%**

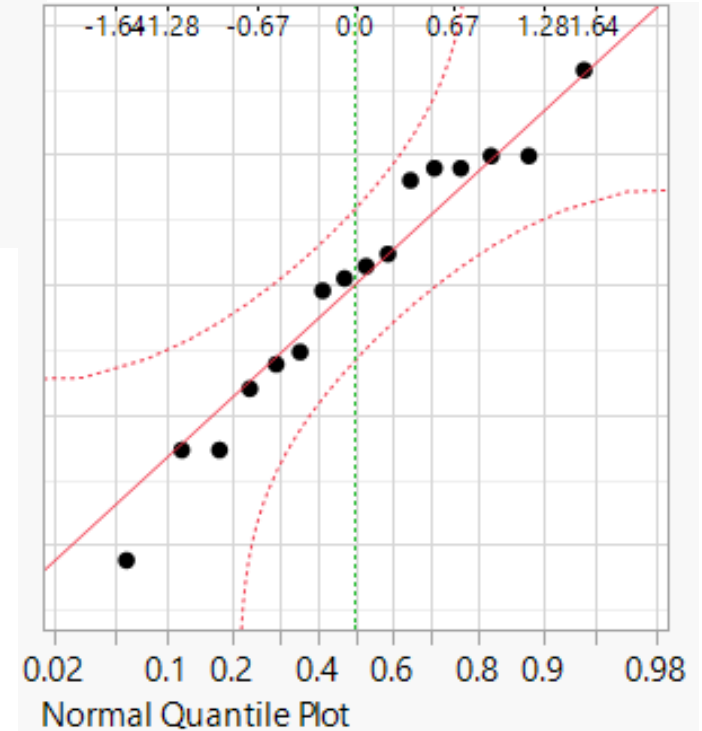
Studentized Resid Yield vs. Predicted Yield



Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.9583521	0.6320
	A ²	Simulated p-Value
Anderson-Darling	0.3137292	0.5252

Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.



All other checks
are ok

The One-Quarter Fraction of 2^k

- Let's consider a screening experiment with $k=6$ factors
- We are interested in **main effects**
- The full experiment has $2^k=64$ treatments
- If we use the one-half fraction $I=ABCDEF$ the resolution is **R=VI**
- **Main effects** are not confounded with $2fi$, $3fi$, $4fi$,
- **Could we use a cheaper fraction?**

The One-Quarter Fraction of $2^k \rightarrow 2^{k-2}$

We use **two generators**

Example with $k=6$

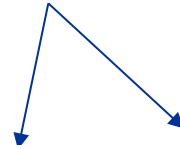
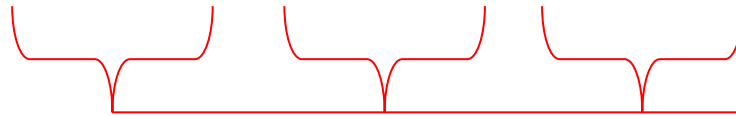


Table 8-9 Construction of the 2^{6-2}_{IV} Design with the Generators $I = ABCE$ and $I = BCDF$

StdOrd	Basic Design				$E = ABC$	$F = BCD$
	A	B	C	D		
1	—	—	—	—	—	—
2	+	—	—	—	+	—
3	—	+	—	—	+	+
4	+	+	—	—	—	+
5	—	—	+	—	+	+
6	+	—	+	—	—	+
7	—	+	+	—	—	—
8	+	+	+	—	+	—
9	—	—	—	+	—	+
10	+	—	—	+	+	+
11	—	+	—	+	+	—
12	+	+	—	+	—	—
13	—	—	+	+	+	—
14	+	—	+	+	—	—
15	—	+	+	+	—	+
16	+	+	+	+	+	+

- The complete **Defining Relation** is obtained by multiplying the Generators **E=ABC** **F=BCD**

$$I = ABCE = BCDF = ADEF$$



Words,
(strings)

- Design Resolution**: minimum length of words belonging to the defining relation (in our case it is IV)
- The **Alias Structure** or **Confounding Structure** of an Effect is produced by the multiplication of that effect by each word in the defining relation

- **Alias structure (or Confounding Structure)**

Table 8-8 Alias Structure for the 2_{IV}^{6-2} Design with $I = ABCE = BCDF = ADEF$

$A = BCE = DEF = ABCDF$	$AB = CE = ACDF = BDEF$
$B = ACE = CDF = ABDEF$	$AC = BE = ABDF = CDEF$
$C = ABE = BDF = ACDEF$	$AD = EF = BCDE = ABCF$
$D = BCF = AEF = ABCDE$	$AE = BC = DF = ABCDEF$
$E = ABC = ADF = BCDEF$	$AF = DE = BCEF = ABCD$
$F = BCD = ADE = ABCEF$	$BD = CF = ACDE = ABEF$
	$BF = CD = ACEF = ABDE$
$ABD = CDE = ACF = BEF$	
$ACD = BDE = ABF = CEF$	

- The four fractions are $E = \pm ABC$, $F = \pm BCD$

Example 8-4 One-Quarter Fraction 2^{6-2}

- An item is produced by the **Injection moulding** process. The post injection **shrinkage** is critical in terms of dimensional tolerances
- The Quality improvement team decided to investigate the effect of 6 Factors on a **critical dimension**.
- The **6 Factors** were:
 - A Mold temperature
 - B Screw speed
 - C Holding time
 - D Cycle time
 - E Gate size
 - F Hold pressure



- To reduce the cost of the experiment, the Quality Improvement Team decided to run **16 treatments** in other words the One-Quarter Fraction of the complete design: **2^{6-2}**
- they used two generators: **$E=ABC$ $F=BCD$**
- the **Defining Relation** is **$I=ABCE=BCDF=ADEF$**

Design characteristics

Factors: 6 **Resolution:** IV

Runs: 16 **Replicates:** 1 **Fraction:** 1/4

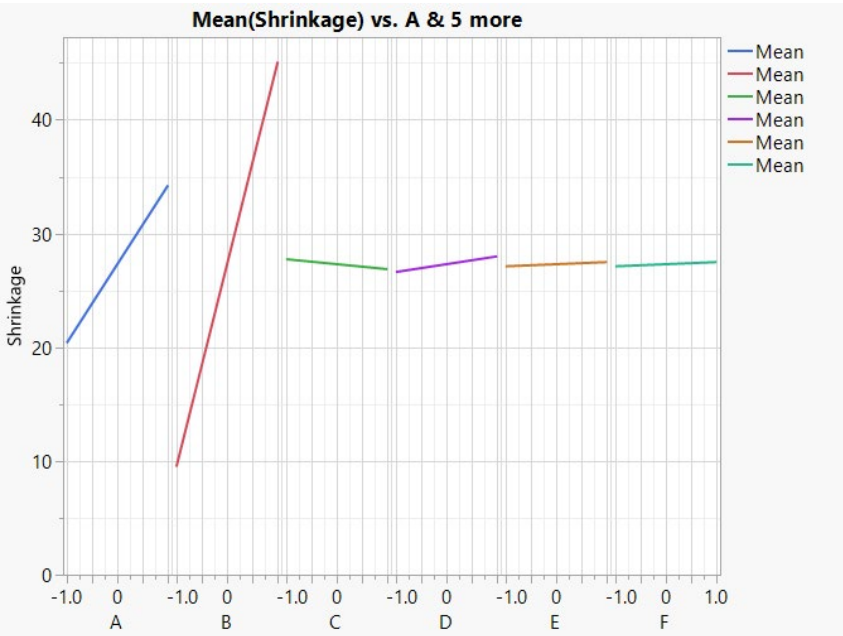
Blocks: none **Center pts (total):** 0

Design Generators: $E = ABC$ $F = BCD$

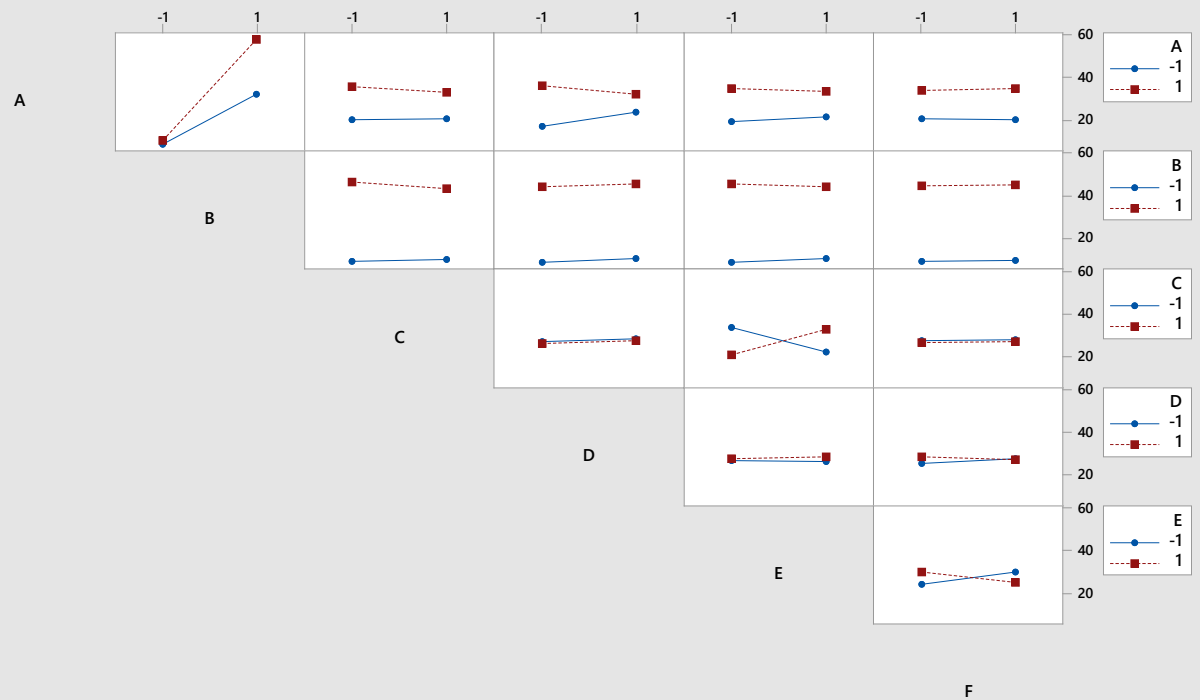
Alias Structure

$I + ABCE + ADEF + BCDF$

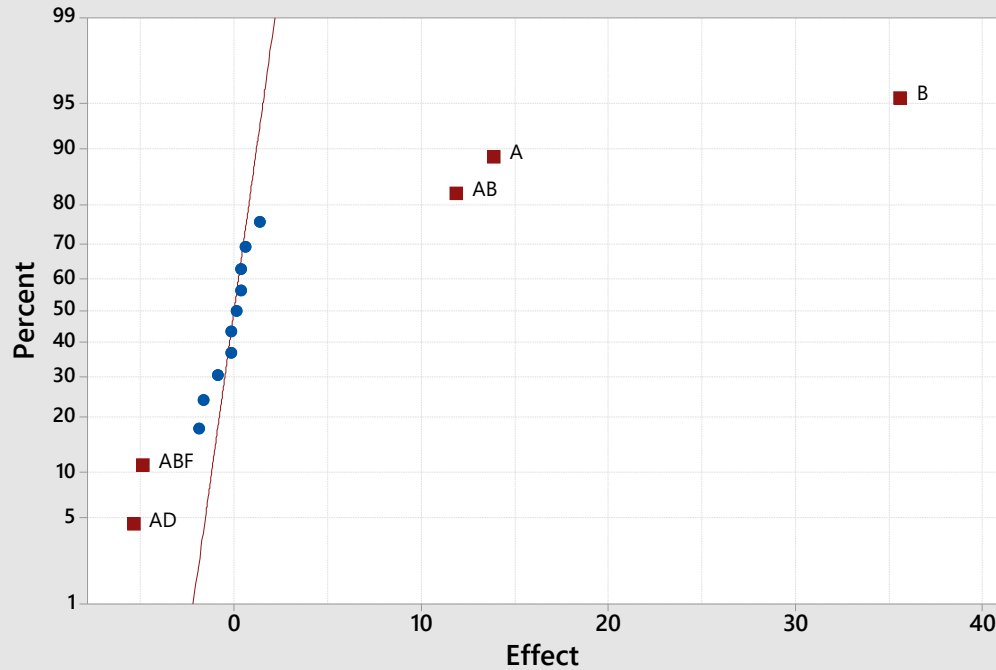
$A + BCE + DEF + ABCDF$
 $B + ACE + CDF + ABDEF$
 $C + ABE + BDF + ACDEF$
 $D + AEF + BCF + ABCDE$
 $E + ABC + ADF + BCDEF$
 $F + ADE + BCD + ABCEF$
 $AB + CE + ACDF + BDEF$
 $AC + BE + ABDF + CDEF$
 $AD + EF + ABCF + BCDE$
 $AE + BC + DF + ABCDEF$
 $AF + DE + ABCD + BCEF$
 $BD + CF + ABEF + ACDE$
 $BF + CD + ABDE + ACEF$
 $ABD + ACF + BEF + CDE$
 $ABF + ACD + BDE + CEF$



Interaction Plot for Shrinkage Data Means



Normal Plot of the Effects
(response is Shrinkage; $\alpha = 0,05$)



Lenth's PSE = 0,9375

Effect Screening

The parameter estimates have equal variances.
The parameter estimates are not correlated.

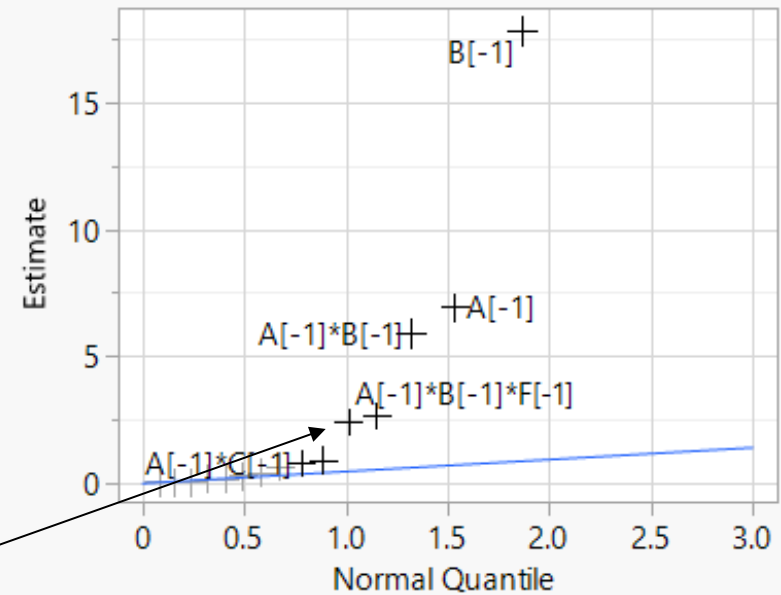
Lenth PSE

0.46875

Orthog t Test used Pseudo Standard Error

Normal Plot

Half Normal Plot ▾



Blue line has slope equal to Lenth's PSE.

AD

- Estimated Effects and Coefficients for shrinkage in coded units**

Term	Effect	Coef
Overall average		27.313
Mold temperature A	13.875	6.937
Screw speed B	35.625	17.812
Holding time C	-0.875	-0.437
Cycle time D	1.375	0.688
Gate size E	0.375	0.187
Hold pressure F	0.375	0.187
temp*speed AB	11,875	5.938
temp*holding time	-1.625	-0.813
temp*cycle time AD	-5.375	-2.688
temp*gate size	-1.875	-0.937
temp*pressure	0.625	0.313
speed*tempo di	-0.125	-0.062
speed*pressure	-0.125	-0.063
temperat*speed*holding time	0.125	0.062
temperat*speed*pressure ABF	-4.875	-2.437

Reduced Model Analysis A, B, AB (Montgomery)

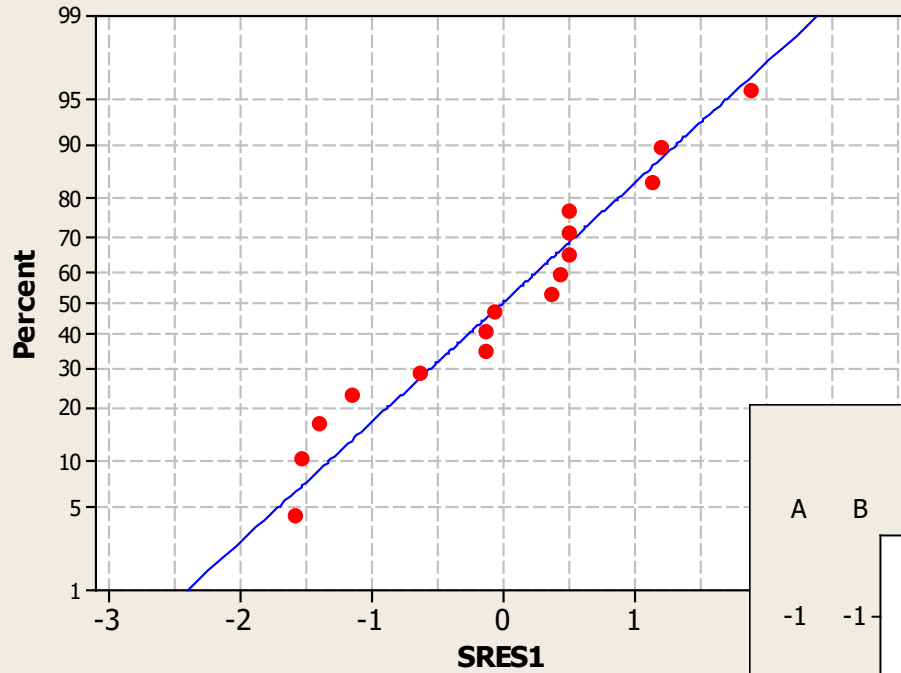
Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	3	6410.7	2136.90	103.09	0.000
Linear	2	5846.6	2923.31	141.02	0.000
A	1	770.1	770.06	37.15	0.000
B	1	5076.6	5076.56	244.90	0.000
2-Way Interactions	1	564.1	564.06	27.21	0.000
A*B	1	564.1	564.06	27.21	0.000
Error	12	248.8	20.73		
Total	15	6659.4			

S	R-sq	R-sq(adj)
4.55293	96.26%	95.33%

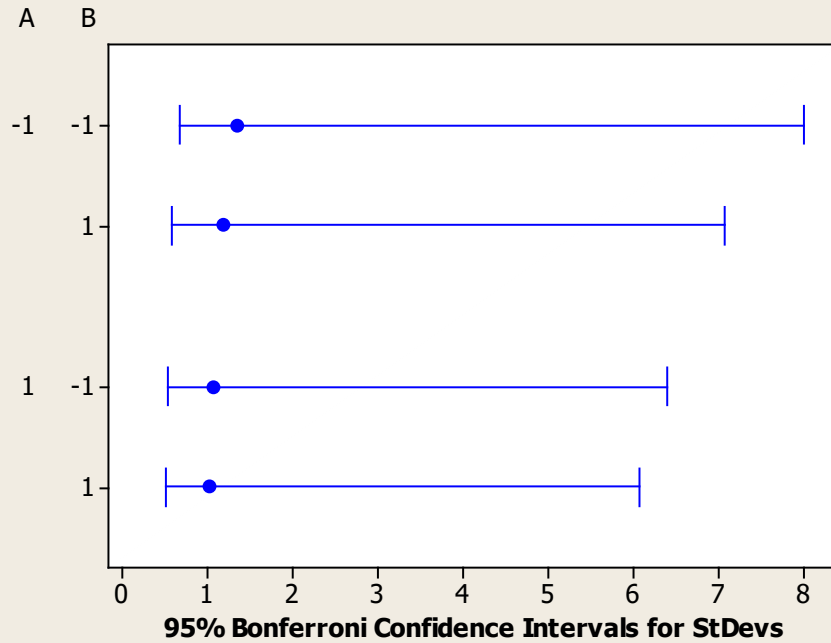
Probability Plot of SRES1

Normal



Mean	-6.52256E-16
StDev	1.033
N	16
AD	0.377
P-Value	0.366

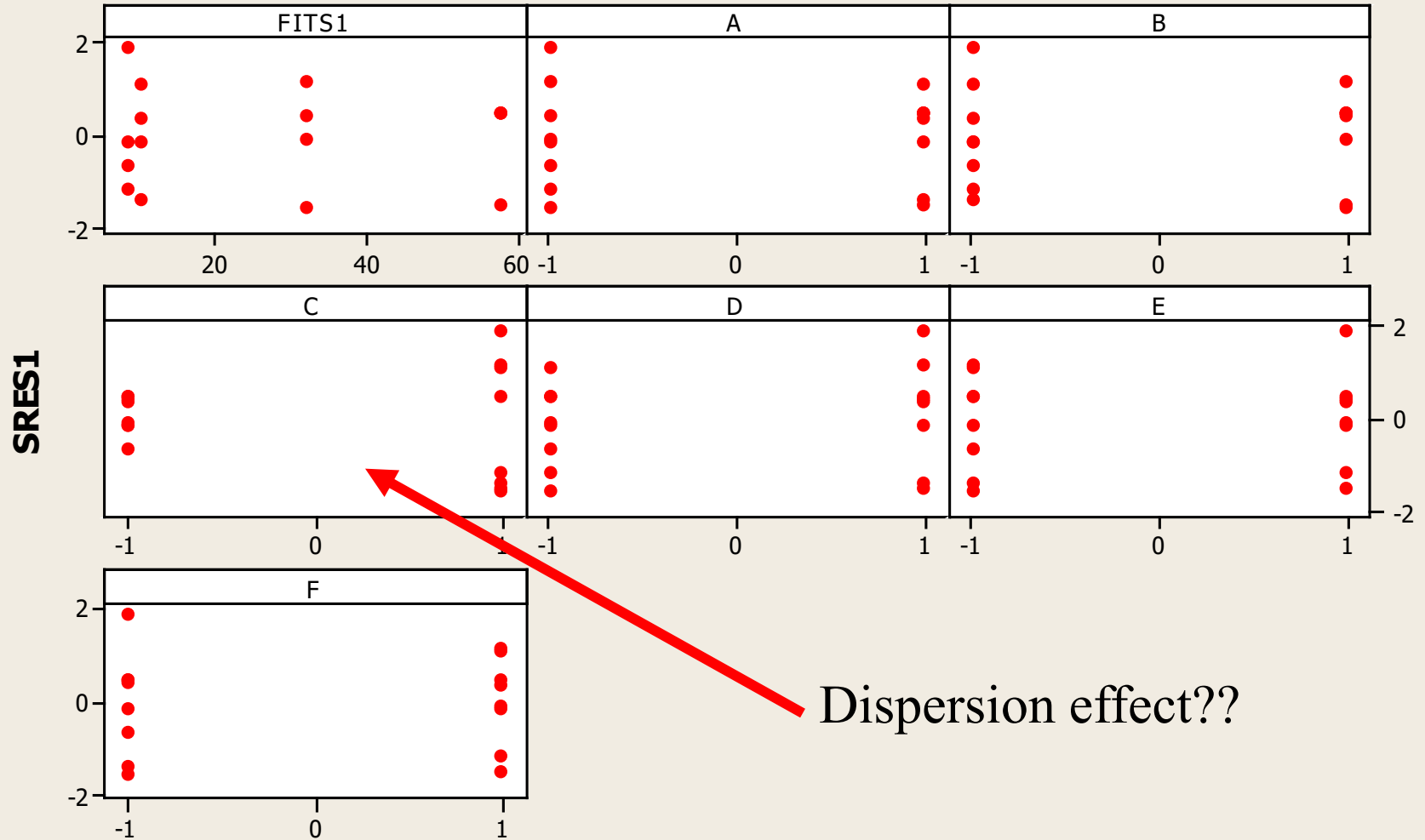
Test for Equal Variances for SRES1



Bartlett's Test	
Test Statistic	0.24
P-Value	0.972
Levene's Test	
Test Statistic	0.16
P-Value	0.924

Residual plots

Scatterplot of SRES1 vs FITS1, A, B, C, D, E, F



Different analysis (qs)

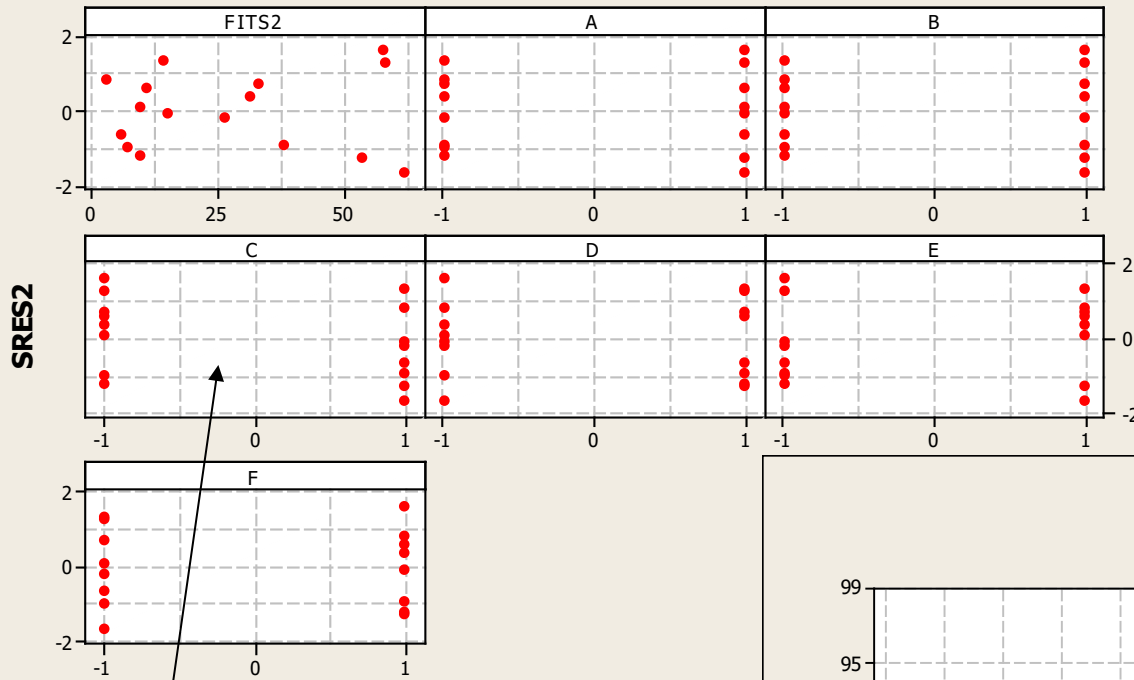
Factorial Regression: Shrinkage versus A; B; D; F

Analysis of Variance.

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	7	6629.44	947.06	252.55	0.000
Linear	4	5854.75	1463.69	390.32	0.000
A	1	770.06	770.06	205.35	0.000
B	1	5076.56	5076.56	1353.75	0.000
(D)	1	7.56	7.56	2.02	0.193
(F)	1	0.56	0.56	0.15	0.709
2-Way Interactions	2	679.62	339.81	90.62	0.000
A*B	1	564.06	564.06	150.42	0.000
A*D	1	115.56	115.56	30.82	0.001
Interactions	1	95.06	95.06	25.35	0.001
A*B*F	1	95.06	95.06	25.35	0.001
Error	8	30.00	3.75		
Total	15	6659.44			
S	R-sq	R-sq(adj)	From	4.553	95.33%
1.93649	99.55%	99.16%	To	1.937	99.16%

QS-AA25-26

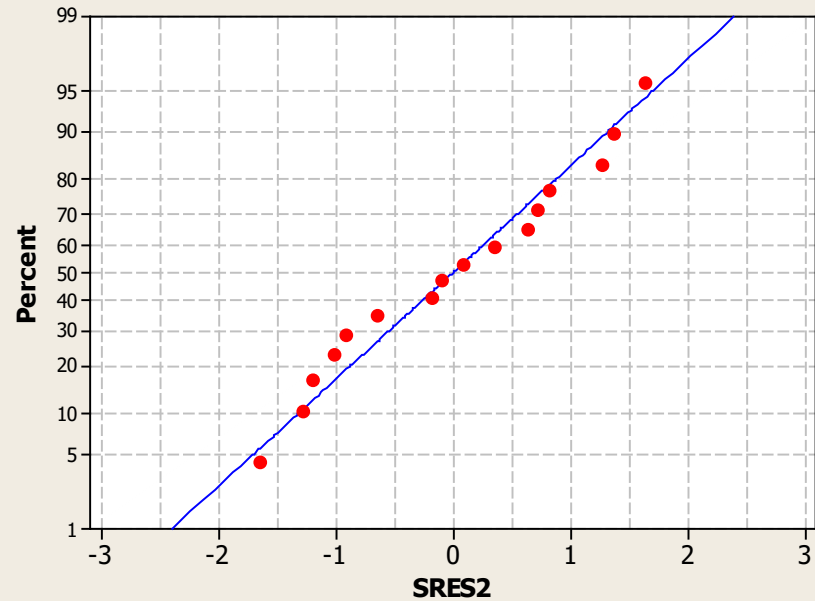
Scatterplot of SRES2 vs FITS2; A; B; C; D; E; F



NO dispersion Effect

Probability Plot of SRES2

Normal



Mean	-2,85882E-15
StDev	1,033
N	16
AD	0,257
P-Value	0,673

Summing up Exercise 8-4

- Injection moulding process with **six** factors
- Compute **Effects**, **Effect Normal probability plot with the Lenth's method**
- According to Montgomery, the model is **A, B, AB** and we note that there could be a possible **Dispersion Effect** regarding factor **C**
- **According to QS, the model is A, B, (D), (F), AB, AD, ABF without any Dispersion Effect but with a more complex model**

The General 2^{k-p} Fractional Factorial Design

- 2^{k-1} = One-Half,
- 2^{k-2} = A Quarter,
- 2^{k-3} = One-Eighth, ...
- **$2^{k-p} = 1/2^p$ fraction**
- Add p columns to the **Basic design** ($k-p$ Factors) using p **independent Generators**
- Each Effect has $2^p - 1$ aliases
- We choose the **Generators** to **maximize** the **Resolution** (Table 8-14 or JMP/Minitab)

Example with $p=2$

- If we choose the following Generators (for 2^{6-2})

$$E=ABC$$

$$I=ABCE$$

$$F=BCD$$

$$I=BCDF$$

- The Defining relation is

$$I=ABCE=BCDF=ADEF \quad (\text{Resolution IV})$$

- **But if we had chosen:**

$$E=ABC$$

$$I=ABCE$$

$$F=ABCD$$

$$I=ABCDF$$

- The Defining relation would have been

$$I=ABCE=ABCDF=DEF \quad (\text{Resolution III})$$

Resolution may not be sufficient

Among the designs with the same Resolution, we select the one which minimizes the number of words with minimum length in the defining relation

Minimum Aberration Design

Table 8-13 Three Choices of Generators for the 2_{IV}^{7-2} Design

Design A Generators: $F = ABC, G = BCD$ $I = ABCF = BCDG = ADFG$	Design B Generators: $F = ABC, G = ADE$ $I = ABCF = ADEG = BCDEFG$	Design C Generators: $F = ABCD, G = ABDE$ $I = ABCDF = ABDEG = CEFG$
Aliases (Two-Factor Interactions)	Aliases (Two-Factor Interactions)	Aliases (Two-Factor Interactions)
$AB = CF$	$AB = CF$	$CE = FG$
$AC = BF$	$AC = BF$	$CF = EG$
$AD = FG$	$AD = EG$	$CG = EF$
$AG = DF$	$AE = DG$	
$BD = CG$	$AF = BC$	
$BG = CD$	$AG = DE$	
$AF = BC = DG$		

Example 8.5

We do not always find what we want!

- Let us consider a **7 factors design**.
- We are interested in estimating the **7 Main Effects** and the **$2fi$** . We assume that the other terms of the model are negligible or of no interest.
- We have to select a design with this alias structure **$me+3fi+\dots$ and $2fi+3fi+\dots \rightarrow$ Resolution V design**

Table 8-14 Selected 2^{k-p} Fractional Factorial Designs

Number of Factors, k	Fraction	Number of Runs	Design Generators
3	2_{III}^{3-1}	4	$C = \pm AB$
4	2_{IV}^{4-1}	8	$D = \pm ABC$
5	2_V^{5-1}	16	$E = \pm ABCD$
	2_{III}^{5-2}	8	$D = \pm AB$ $E = \pm AC$
6	2_{VI}^{6-1}	32	$F = \pm ABCDE$
	2_{IV}^{6-2}	16	$E = \pm ABC$ $F = \pm BCD$
	2_{III}^{6-3}	8	$D = \pm AB$ $E = \pm AC$ $F = \pm BC$
7	2_{VII}^{7-1}	64	$G = \pm ABCDEF$
	2_{IV}^{7-2}	32	$F = \pm ABCD$ $G = \pm ABDE$
	2_{IV}^{7-3}	16	$E = \pm ABC$ $F = \pm BCD$ $G = \pm ACD$
	2_{III}^{7-4}	8	$D = \pm AB$ $E = \pm AC$ $F = \pm BC$ $G = \pm ABC$

In table 8.14 we find

A

$$2_{VII}^{7-1} \quad N=64$$

B

$$2_{IV}^{7-2} \quad N=32$$

C

$$2_{IV}^{7-3} \quad N=16$$

Option A

2_{VII}^{7-1} 64 runs

It is very far from our target

Option B

2_{IV}^{7-2} 32 runs

Generators:

– **F=ABCD**

I=ABCDF

– **G=ABDE**

I=ABDEG

Defining Relation $I=ABCDF=ABDEG=CEFG$

Alias Structure

$A=BCDF=BDEG=AGEGF$ B=....

$AB=CDF=DEG=ABCEFG$

$CE=FG=ABEDF=ABCDG$

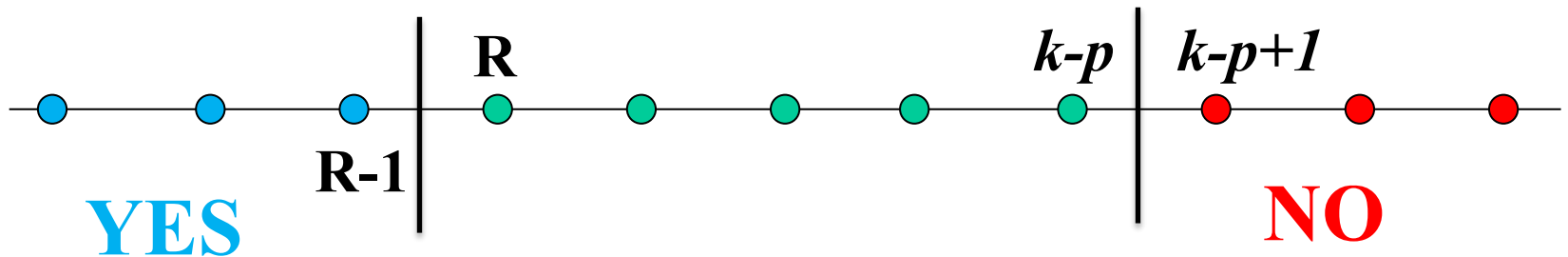
$CF=EG=ABD= ABCDEF$ etc etc

Option C 2_{IV}^{7-3} 16 runs

- **Generators:** $E=ABC$ $F=BCD$ $G=ACD$
- **Defining Relation**
 $I=ABEC=BCDF=ACDG=ADEF=BDEG=ABFG$
 $=CEFG$
- The alias structure is very intricate (see table XII and **check by hand**)
- **Conclusion:** I prefer solution B

Projection of the 2_R^{k-p} Fractional Factorial

- How many Factors (subset of the original Factors) collapse in a full design?
- **upper bound:** a subset of more than $(k-p)$ factors does not project in a full design (we don't have enough data).
- **lower bound:** a subset of factors $\leq (R-1)$ projects in full designs
- The subset of factors belonging to the interval $[R ; (k-p)]$ collapses in **full** designs for subset of factors which are not in the **Defining relation** or are not derivable from them



Note that $(k-p)$ and R could be reversed: $2^{(7-1)}$ has resolution **VII**

Example 1 $k=6$ $p=2$ $N=16$

Defining relation $I=ABCE=BCDF=ADEF$

$$R=IV \quad (k-p)=4 \quad \rightarrow 2_{IV}^{6-2}$$

- The projections in subgroups of original factors up to $(R-1)=3$ factors are **Full Factorial Designs**
- greater or equal $(k-p)+1=5 \rightarrow$ **No subset**
- Each subset of size 4 which is not a “word” of the defining relation produces a full factorial design
 - A,B,C,E (one-half fraction replicated)
 - A,D,E,F (one-half fraction replicated)
 - B,C,D,F (one-half fraction replicated)
 - A,B,C,F (full factorial)
 - A,B,C,D (full factorial)
 -

The subset **A, B, C, E** belongs to the defining relation (ABCE)

Table 8-9 Construction of the 2_{IV}^{6-2} Design with the Generators $I = ABCE$ and $I = BCDF$

Run	Basic Design				$E = ABC$	$F = BCD$
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>		
1	—	—	—	—	—	—
2	+	—	—	—	+	—
3	—	+	—	—	+	+
4	+	+	—	—	—	+
5	—	—	+	—	+	+
6	+	—	+	—	—	+
7	—	+	+	—	—	—
8	+	+	+	—	+	—
9	—	—	—	+	—	+
10	+	—	—	+	+	+
11	—	+	—	+	+	—
12	+	+	—	+	—	—
13	—	—	+	+	+	—
14	+	—	+	+	—	—
15	—	+	+	+	—	+
16	+	+	+	+	+	+

One-half design, replicated

- **Example 2**
- **k=7 p=2 k-p=5 N=32**
- Let us use the generators **F=ABCD G=ABDE**
- Defining relation **I=ABCDF=ABDEG=CEFG**
- Resolution **R=IV**
- **R-1=3** **k-p+1=6** **uncertain 4 and 5**
- **Full design in:**
 - All the subsets up to **three** Factors
 - All the subsets of **four** Factors excluding (C,E,F,G)
 - All the subsets of **five** Factors excluding (A,B,C,D,F), (A,B,D,E,G) and all the (C,E,F,G,*)

Resolution III Designs

- In **Resolution III** Designs **Main Effects** are **aliased** with $2fi$ and higher Factor Interactions.
- Used for **screening** experiments with minimum number of experiments (i.e. 5 – 7 factors with 8 runs, 9 - 15 factors with 16 runs)
- I suggest you **not to use** them

Create Factorial Design: Display Available Designs

Available Factorial Designs (with Resolution)

	Factors														
Run	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
4	Full	III													
8		Full	IV	III	III	III									
16			Full	V	IV	IV	IV	III	III	III	III	III	III	III	
32				Full	VI	IV	IV	IV	IV	IV	IV	IV	IV	IV	
64					Full	VII	V	IV	IV	IV	IV	IV	IV	IV	
128						Full	VIII	VI	V	V	IV	IV	IV	IV	

Available Resolution III Plackett-Burman Designs

Factors	Runs	Factors	Runs	Factors	Runs
2-7	12,20,24,28,...,48	20-23	24,28,32,36,...,48	36-39	40,44,48
8-11	12,20,24,28,...,48	24-27	28,32,36,40,44,48	40-43	44,48
12-15	20,24,28,36,...,48	28-31	32,36,40,44,48	44-47	48
16-19	20,24,28,32,...,48	32-35	36,40,44,48		

Help

OK

Fold over

- **Fold over** is an experimentation technique based on a sequence of experiments.
- The goal is to assemble fractions to **clear up** ambiguities of interest in the alias structure

Let us run the principal fraction

Table 8-19 The 2^{7-4}_{III} Design with the Generators $I = ABD$, $I = ACE$, $I = BCF$, and $I = ABCG$

	Basic Design			$D = AB$	$E = AC$	$F = BC$	$G = ABC$	
	A	B	C					
1	−	−	−	+	+	+	−	<i>def</i>
2	+	−	−	−	−	+	+	<i>afg</i>
3	−	+	−	−	+	−	+	<i>beg</i>
4	+	+	−	+	−	−	−	<i>abd</i>
5	−	−	+	+	−	−	+	<i>cdg</i>
6	+	−	+	−	+	−	−	<i>ace</i>
7	−	+	+	−	−	+	−	<i>bcf</i>
8	+	+	+	+	+	+	+	<i>abcdefg</i>

Aliasing of Effects

Effects	Aliases
A	= B*D = C*E = F*G
B	= A*D = C*F = E*G
C	= A*E = B*F = D*G
D	= A*B = C*G = E*F
E	= A*C = B*G = D*F
F	= A*G = B*C = D*E
G	= A*F = B*E = C*D

The alias table for the first fraction (limited to 2fi)

First Fraction Complete Defining relationship

$I=ABD=ACE=BCF=ABCG=BCDE=ACDF=CDG=ABEF=BEG=$
 $=AFG=DEF=ADEG=BDGF=CEFG=ABCDEFG$

JMP Output you can specify up to what order to report the interactions

$C=1\ 2\ 4 = 1\ 3\ 5 = 2\ 3\ 4\ 5 = 2\ 3\ 6 = 1\ 3\ 4\ 6 = 1\ 2\ 5\ 6 = 4\ 5\ 6 = 1\ 2\ 3$
 $7 = 3\ 4\ 7 = 2\ 5\ 7 = 1\ 4\ 5\ 7 = 1\ 6\ 7 = 2\ 4\ 6\ 7 = 3\ 5\ 6\ 7 = 1\ 2\ 3\ 4\ 5\ 6\ 7$

Run a **Second fraction**:

we switch the signs of the **D** column i.e. we change it to $(- + + - - + + -) \rightarrow \mathbf{D = -AB}$

Second Fraction Defining relationship

$\mathbf{I = -ABD = ACE = BCF = ABCG = -BCDE = -ACDF = CDG = AB EF = BEG = AFG = -DEF = -ADEG = -BDGF = CEFG = -ABCDEFG}$

Aliasing of Effects

Effects	Aliases
A	$= -B*D = C*E = F*G$
B	$= -A*D = C*F = E*G$
C	$= A*E = B*F = -D*G$
D	$= -A*B = -C*G = -E*F$
E	$= A*C = B*G = -D*F$
F	$= A*G = B*C = -D*E$
G	$= A*F = B*E = -C*D$

We can compute l from the first fraction and l^* from the second fraction

If we assemble the two estimates, l and l^* , we have:

	$(l + l^*)/2$	$(l - l^*)/2$
$- l^*_A = A - BD + CE + FG$	$A + CE + FG$	BD
$- l^*_B = B - AD + CF + EG$	$B + CF + EG$	AD
$- l^*_D = D - AB - CG - EF$	D	$AB + CG + EF$
$- l^*_C = C + AE + BF - DG$	$C + AE + BF$	DG
$- l^*_E = E + AC + BG - DF$	$E + AC + BG$	DF
$- l^*_F = F + BC + AG - DE$	$F + BC + AG$	DE
$- l^*_G = G - CD + BE + AF$	$G + BE + AF$	CD

- Remark:** interaction terms higher than two are not shown

By changing the D signs we have separated *me* D from its 2fi

Combined experiment

First fraction: $I=ABD$ $I=ACE$ $I=BCF$ $I=ABCG$

Second fraction: $I=-ABD$ $I=ACE$ $I=BCF$ $I=ABCG$

L Number of equal generators (Like) in the two fractions: **3**

U Number of unequal generators (Unlike) in the two fractions: **1**

The number of generators of the **Joint design** is

$L+U-1=3+1-1=3$ (remember that now we have 16 runs $\rightarrow 2^{7-3}$)

1) Take the 3 Like generators

2) Add the product of the Unlike generators (if more than one); in this case the unique Unlike generator **do not contribute** to the defining relation

As usual, to obtain the defining relation you must multiply the defining relations **I=ACE I=BCF I=ABCG** **Resolution III**

$$\mathbf{C = 1\ 3\ 5 = 2\ 3\ 6 = 1\ 2\ 5\ 6 = 1\ 2\ 3\ 7 = 2\ 5\ 7 = 1\ 6\ 7 = 3\ 5\ 6\ 7}$$

Alias Structure

$$\mathbf{I + ACE + AFG + BCF + BEG + ABCG + ABEF + CEFG}$$

$$\mathbf{A + CE + FG + BCG + BEF + ABCF + ABEG + ACEFG}$$

$$\mathbf{B + CF + EG + ACG + AEF + ABCE + ABFG + BCEFG}$$

$$\mathbf{C + AE + BF + ABG + EFG + ACFG + BCEG + ABCEF}$$

$$\mathbf{D + ACDE + ADFG + BCDF + BDEG + ABCDG + ABDEF + CDEFG}$$

$$\mathbf{E + AC + BG + ABF + CFG + ACFG + BCEF + ABCEG}$$

$$\mathbf{F + AG + BC + ABE + CEG + ACEF + BEFG + ABCFG}$$

$$\mathbf{G + AF + BE + ABC + CEF + ACEG + BCFG + ABEFG}$$

$$\mathbf{AB + CG + EF + ACF + AEG + BCE + BFG + ABCEFG}$$

$$\mathbf{AD + CDE + DFG + BCDG + BDEF + ABCDF + ABDEG + ACDEFG}$$

$$\mathbf{BD + CDF + DEG + ACDG + ADEF + ABCDE + ABDFG + BCDEFG}$$

$$\mathbf{CD + ADE + BDF + ABDG + DEFG + ACDFG + BCDEG + ABCDEF}$$

$$\mathbf{DE + ACD + BDG + ABDF + CDFG + ADEFG + BCDEF + ABCDEG}$$

$$\mathbf{DF + ADG + BCD + ABDE + CDEG + ACDEF + BDEFG + ABCDFG}$$

$$\mathbf{DG + ADF + BDE + ABCD + CDEF + ACDEG + BCDFG + ABDEFG}$$

$$\mathbf{ABD + CDG + DEF + ACDF + ADEG + BCDE + BDFG + ABCDEFG}$$

From MINITAB

Full Fold-Over Design

Switch the signs in **all** factor columns and de-aliases all Main Effects from their two-factor interaction alias chains .

1° Fraction $I=ABD$ $I=ACE$ $I=BCF$ $I=ABCG$

2° Fraction $I=-ABD$ $I=-ACE$ $I=-BCF$ $I=ABCG$

Defining Relation for a **Full Fold-Over design**

L Number of equal generators (Like) in the 2 Fractions -> 1

U Number of unequal generators (Unlike) in the 2 Fractions-> 3

The number of generators of the joint design is

$$\mathbf{L+U-1} = 1+3-1=3$$

The first one is the Like generator $\mathbf{I=ABCG}$.

For the others, take the product of two **Unlike generators**

$$\mathbf{I=ABD*ACE=BCDE} \quad \mathbf{I=ABD*BCF=ACDF}$$

Resolution IV

Alias Structure

I + ABCG + ABEF + ACDF + ADEG + BCDE + BDFG + CEFG

A + BCG + BEF + CDF + DEG + ABCDE + ABDFG + ACEFG

B + ACG + AEF + CDE + DFG + ABCDF + ABDEG + BCEFG

C + ABG + ADF + BDE + EFG + ABCEF + ACDEG + BCDFG

D + ACF + AEG + BCE + BFG + ABCDG + ABDEF + CDEFG

E + ABF + ADG + BCD + CFG + ABCEG + ACDEF + BDEFG

F + ABE + ACD + BDG + CEG + ABCFG + ADEFG + BCDEF

G + ABC + ADE + BDF + CEF + ABEFG + ACDFG + BCDEG

AB + CG + EF + ACDE + ADFG + BCDF + BDEG + ABCEFG

AC + BG + DF + ABDE + ACFG + BCEF + CDEG + ABCDFG

AD + CF + EG + ABCE + ABFG + BCDG + BDEF + ACDEFG

AE + BF + DG + ABCD + ACFG + BCEG + CDEF + ABDEFG

AF + BE + CD + ABDG + ACEG + BCFG + DEFG + ABCDEF

AG + BC + DE + ABDF + ACEF + BEFG + CDFG + ABCDEG

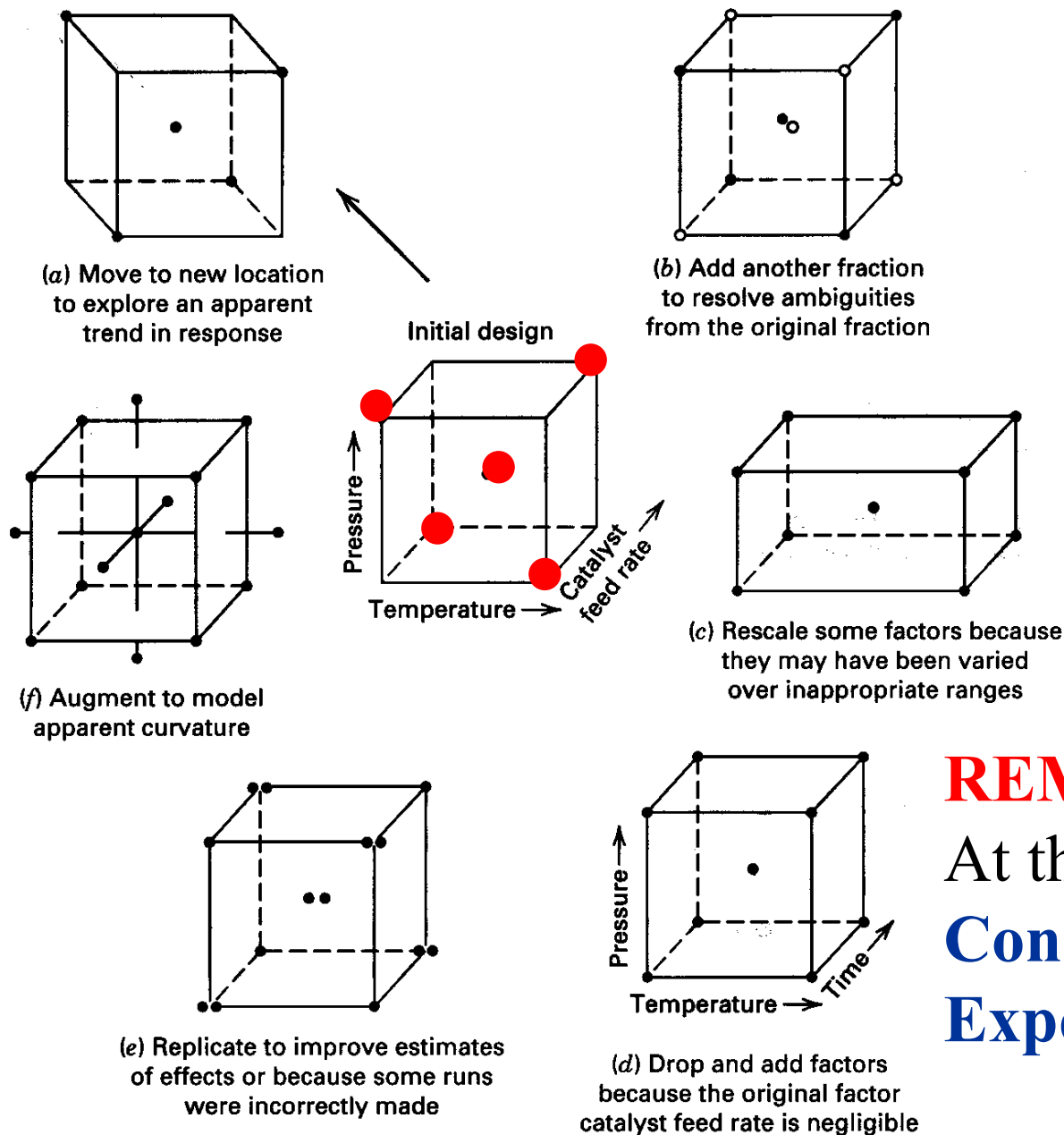
BD + CE + FG + ABCF + ABEG + ACDG + ADEF + BCDEFG

ABD + ACE + AFG + BCF + BEG + CDG + DEF + ABCDEFG

Resolution IV Designs 2_{IV}^{k-p}

- In **Resolution IV Design**:
 - **Main Effects** are **not confused** with $2fi$
 - $2fi$ are **confused** with $2fi$
- If the design have **N=2k** treatments, the design is called **Minimum**
- **I suggest** you to use **Resolution IV (or higher Resolution) Designs**
- There are special rules for Resolution IV designs, and other methods for adding runs to fractions to de-alias Effects of interest .

General approach to **Sequential Experimentation**



REMARK:

At the end of the study:
**Confirmation
Experiments**

Figure 8-11 Possibilities for follow-up experimentation after a fractional factorial experiment (adapted from Box (1992–1993), with permission of the publisher).